Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Heat and material balance, line sizing and pump hydraulics

NDFH-V1-PE-HMB-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-PE-HMB-001
Title	Heat and material balance, line sizing and pump hydraulics
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Design, maximum, turndown and hot-day cases of H-100 and E-100; line sizing; P-100A/B hydraulics.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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1 · Design basis and cases

Each case was run on a copy of the H-100 model for 40 min (TIC-030 in AUTO at 350 °C, combustion air trimmed to 2.0 % firebox O_2, trips disabled so the run is not cut short).

Case	Feed, L/s	Fired, kW	Absorbed, kW	Efficiency	Firebox O_2 %	Tube skin, °C
Design case	6.80	2835	2312	81.6 %	2.00	398
Maximum rate (110 %)	7.50	3161	2550	80.7 %	2.00	398
Turndown (5.2 L/s)	5.20	2113	1768	83.7 %	2.00	397
Design rate, hot day (40 °C)	6.80	2804	2312	82.5 %	2.00	398

Assumptions used in this document

No.	Assumed	Value	Used in
A1	Feed viscosity	1.0 mPa s at 40 °C, 0.5 mPa s at 180–350 °C (Design tab value)	Line sizing
A2	Feed vapour pressure at 40 °C	5 kPa(a)	NPSH available
A3	Pump efficiency P-100A/B	60 % at rated flow	Absorbed power, motor size
A4	Heating fluid from Unit A	hot oil, ρ 850 kg/m³, c_p 2.3 kJ/kg K, return 150 °C, operating 600 kPa(g), design 1000 kPa(g) / 260 °C	E-100 rating, tube-rupture case
A5	E-100 overall U	350 W/m² K (liquid–liquid, clean), 15 % area margin	E-100 area
A6	E-100 geometry	TEMA AES, ¾ in 14 BWG tubes (ID 14.8 mm), shell OD 0.61 m × 6.1 m	Wetted area, rupture area
A7	Fuel oil	ρ 900 kg/m³, μ 10 mPa s, delivered heated to 120 °C, flash point 66 °C	Fuel line, area classification
A8	Feed fluid	flash point 45 °C, auto-ignition 250 °C, latent heat 300 kJ/kg, vapour M 150, k 1.05 at relief	Relief and area classification
A9	Allowable stresses	illustrative ASME II-D values (see the stress table); confirm from the code edition in force	Wall thickness
A10	Electrical	400 V, 3-phase, 50 Hz; motor η 92 %, pf 0.86; starting current 6 × FLC	Load list, cable sizing
A11	Lead times	typical market lead times in weeks, for teaching	Procurement schedule
A12	Corrosion allowance	3 mm carbon steel coil and piping, 3 mm vessels	Wall thickness

2 · Heat and material balance: Design case

Stream	Description	Phase	Mass, kg/h	Volume, m³/h	T, °C	P, kPa(g)
1	Feed from TK-101	Liquid	19584	24.48	40.0	50
2	P-100 discharge	Liquid	19584	24.48	40.0	1030
3	Feed from E-100 to coil	Liquid	19584	24.48	180.0	845
4	Crossover, convection → radiant	Liquid	19584	—	233.1	≈ atm
5	Heater outlet to V-200	Liquid	19584	—	350.0	545
6	Heating fluid from Unit A (A4)	Liquid	37250	43.82	230.0	600
7	Fuel oil to burner (A7)	Liquid	243	0.27	120.0	1940
8	Combustion air	Gas	3867	3212.05	20.0	≈ atm
9	Flue gas leaving firebox	Gas	4110	—	986.8	≈ atm
10	Flue gas to atmosphere	Gas	4355	—	399.3	≈ atm

Duty	E-100	Convection	Radiant	Absorbed	Fired (LHV)	Efficiency	Valves FV-010/FV-300/FV-200/HV-400
kW	1904	722	1590	2312	2835	81.6 %	57 / 45 / 48 / 67 %

Heat and material balance: Maximum rate (110 %)

Stream	Description	Phase	Mass, kg/h	Volume, m³/h	T, °C	P, kPa(g)
1	Feed from TK-101	Liquid	21600	27.00	40.0	50
2	P-100 discharge	Liquid	21600	27.00	40.0	950
3	Feed from E-100 to coil	Liquid	21600	27.00	180.0	790
4	Crossover, convection → radiant	Liquid	21600	—	234.2	≈ atm
5	Heater outlet to V-200	Liquid	21600	—	350.0	425
6	Heating fluid from Unit A (A4)	Liquid	41085	48.33	230.0	600
7	Fuel oil to burner (A7)	Liquid	271	0.30	120.0	1926
8	Combustion air	Gas	4312	3581.07	20.0	≈ atm
9	Flue gas leaving firebox	Gas	4583	—	1013.4	≈ atm
10	Flue gas to atmosphere	Gas	4827	—	419.5	≈ atm

Duty	E-100	Convection	Radiant	Absorbed	Fired (LHV)	Efficiency	Valves FV-010/FV-300/FV-200/HV-400
kW	2100	814	1737	2550	3161	80.7 %	62 / 53 / 56 / 67 %

Heat and material balance: Turndown (5.2 L/s)

Stream	Description	Phase	Mass, kg/h	Volume, m³/h	T, °C	P, kPa(g)
1	Feed from TK-101	Liquid	14976	18.72	40.0	50
2	P-100 discharge	Liquid	14976	18.72	40.0	1184
3	Feed from E-100 to coil	Liquid	14976	18.72	180.0	951
4	Crossover, convection → radiant	Liquid	14976	—	229.9	≈ atm
5	Heater outlet to V-200	Liquid	14976	—	350.0	776
6	Heating fluid from Unit A (A4)	Liquid	28485	33.51	230.0	600
7	Fuel oil to burner (A7)	Liquid	181	0.20	120.0	1967
8	Combustion air	Gas	2882	2393.88	20.0	≈ atm
9	Flue gas leaving firebox	Gas	3063	—	916.9	≈ atm
10	Flue gas to atmosphere	Gas	3308	—	350.3	≈ atm

Duty	E-100	Convection	Radiant	Absorbed	Fired (LHV)	Efficiency	Valves FV-010/FV-300/FV-200/HV-400
kW	1456	519	1249	1768	2113	83.7 %	43 / 31 / 31 / 67 %

Heat and material balance: Design rate, hot day (40 °C)

Stream	Description	Phase	Mass, kg/h	Volume, m³/h	T, °C	P, kPa(g)
1	Feed from TK-101	Liquid	19584	24.48	40.0	50
2	P-100 discharge	Liquid	19584	24.48	40.0	1030
3	Feed from E-100 to coil	Liquid	19584	24.48	180.0	845
4	Crossover, convection → radiant	Liquid	19584	—	232.7	≈ atm
5	Heater outlet to V-200	Liquid	19584	—	350.0	545
6	Heating fluid from Unit A (A4)	Liquid	37250	43.82	230.0	600
7	Fuel oil to burner (A7)	Liquid	240	0.27	120.0	1941
8	Combustion air	Gas	3825	3176.62	40.0	≈ atm
9	Flue gas leaving firebox	Gas	4065	—	987.8	≈ atm
10	Flue gas to atmosphere	Gas	4310	—	398.3	≈ atm

Duty	E-100	Convection	Radiant	Absorbed	Fired (LHV)	Efficiency	Valves FV-010/FV-300/FV-200/HV-400
kW	1904	716	1596	2312	2804	82.5 %	57 / 45 / 51 / 67 %

Overall balances at design

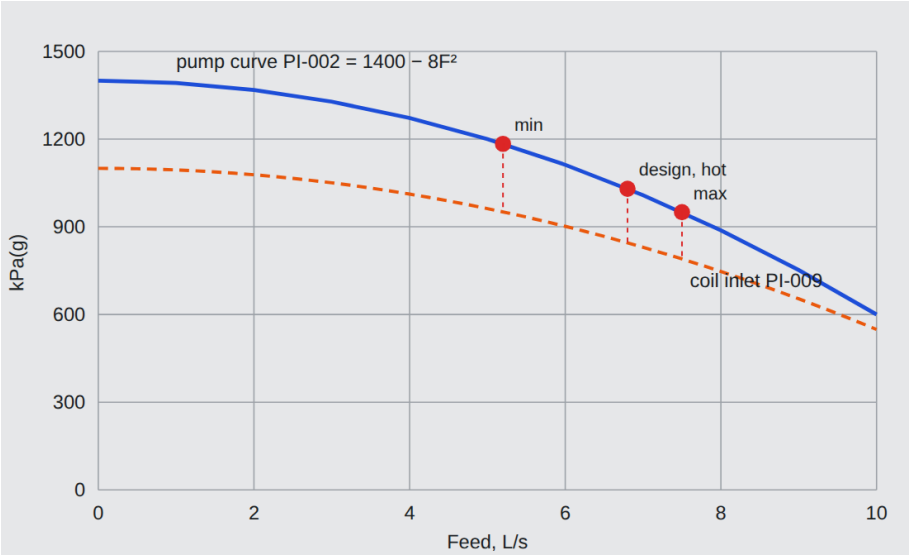
Balance	In	Out	Difference
Feed in = heater outlet	19584 kg/h	19584 kg/h	0.00 %
Air + fuel = flue gas leaving firebox	4110 kg/h	4110 kg/h	0.00 %
Firebox flue + in-leakage = stack	4355 kg/h	4355 kg/h	0.00 %
Radiant + convection = absorbed	2312 kW	2312 kW	0.00 %
Absorbed = $\dot{m}\, c_p\, (T_{out} - T_{in})$	2312 kW	2312 kW	0.00 %

3 · Line sizing

Criteria (typical practice, not a code): pump suction 0.6–1.5 m/s and ≤ 10 kPa per 100 m; liquid discharge 1–3 m/s and ≤ 45 kPa per 100 m. Darcy–Weisbach with the Churchill friction factor, roughness 0.045 mm.

Line	Service	Criterion	Design m³/h	v m/s	ΔP kPa/100 m	Max v m/s	Max ΔP	Result
4"-P-101	Feed, TK-101 to P-100 suction	pump suction: 0.6–1.5 m/s, ≤ 10	24.48	0.83	5.7	0.91	6.8	✓
3"-P-102	Feed, P-100 discharge to E-100	liquid discharge: 1–3 m/s, ≤ 45	24.48	1.43	22.0	1.57	26.4	✓
3"-P-103	Feed, E-100 to coil inlet (FV-010)	liquid discharge: 1–3 m/s, ≤ 45	24.48	1.43	20.4	1.57	24.6	✓
3"-P-105	Heater outlet to V-200	liquid discharge: 1–3 m/s, ≤ 45	24.48	1.43	20.4	1.57	24.6	✓
3"-HF-401	Heating fluid from Unit A to E-100 (A4)	liquid discharge: 1–3 m/s, ≤ 45	43.82	2.55	69.8	2.82	84.2	✗
1"-FO-201	Fuel oil to the burner (A7)	small fuel line (1 in minimum): 0.1–1.5 m/s, ≤ 45	0.27	0.13	6.1	0.15	6.8	✓
Air duct DN350 (TBC)	Combustion air to windbox	air duct ≤ 15 m/s	3212	9.3	—	10.3	—	✓

4 · Pump hydraulics, P-100A/B



Pump curve from the model: PI-002 = 1400 − 8 F² kPa(g). FV-010 takes up the difference between the pump curve and what the coil needs.

Item	Value
Normal / rated flow	6.80 / 7.50 L/s (27.0 m³/h)
Suction PI-001	50 kPa(g)
Discharge at design / rated	1030 / 950 kPa(g)
Differential head at rated	114.7 m
Shut-off	1400 kPa(g), 172 m
Hydraulic / absorbed power at rated	6.75 / 11.2 kW (A3)
Motor	15 kW
NPSH available (A2) / required	18.6 / 3 m
FV-010 pressure drop at design	185 kPa (19 % of 980 kPa); coil 300 kPa

5 · Checks

Check	This document	Compared with	Result
HMB stream 1 flow = FIC-003 at design	24.48 m³/h	24.48 m³/h (model now)	✓
HMB outlet temperature = TIC-030 set point	350.0 °C	350.0 °C	✓
HMB fuel = FI-001 at design	243.0 kg/h	243.3 kg/h	✓
HMB air = FIC-005 at design	3867 kg/h	3892 kg/h	✓
Absorbed duty = Design tab duty	2312 kW	2312 kW (Design tab)	✓
Balance: Feed in = heater outlet	19584 kg/h	19584 kg/h	✓
Balance: Air + fuel = flue gas leaving firebox	4110 kg/h	4110 kg/h	✓
Balance: Firebox flue + in-leakage = stack	4355 kg/h	4355 kg/h	✓
Balance: Radiant + convection = absorbed	2312 kW	2312 kW	✓
Balance: Absorbed = $\dot{m}\, c_p\, (T_{out} - T_{in})$	2312 kW	2312 kW	✓
Pump curve at design = PI-002	1030 kPa(g)	1030 kPa(g)	✓
NPSH available ≥ NPSH required + 1 m	18.6 m (A2)	4.0 m	✓
FV-010 ΔP at design ≥ 70 kPa or 25 % of the system	185 kPa	19 % of 980 kPa	✓
Line 4"-P-101 within its criteria at the maximum case	0.91 m/s, 6.8 kPa/100 m	≤ 1.5 m/s, ≤ 10 kPa/100 m	✓
Line 3"-P-102 within its criteria at the maximum case	1.57 m/s, 26.4 kPa/100 m	≤ 3 m/s, ≤ 45 kPa/100 m	✓
Line 3"-P-103 within its criteria at the maximum case	1.57 m/s, 24.6 kPa/100 m	≤ 3 m/s, ≤ 45 kPa/100 m	✓
Line 3"-P-105 within its criteria at the maximum case	1.57 m/s, 24.6 kPa/100 m	≤ 3 m/s, ≤ 45 kPa/100 m	✓
Line 3"-HF-401 within its criteria at the maximum case	2.82 m/s, 84.2 kPa/100 m	≤ 3 m/s, ≤ 45 kPa/100 m	✗
Line 1"-FO-201 within its criteria at the maximum case	0.15 m/s, 6.8 kPa/100 m	≤ 1.5 m/s, ≤ 45 kPa/100 m	✓